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Profiles of Scientific Thinking

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Identifying how students engage in scientific thinking—especially when learning about socially relevant science topics—is essential for designing instruction that meets diverse learning needs. This study used a person-centered approach to investigate scientific thinking profiles among middle and high school students engaged in scaffolded science lessons. Drawing on 1,191 observations from 197 students in science classrooms in the Mid-Atlantic and Southeast United States, we employed latent profile analysis, longitudinal latent profile analysis, and latent transition analysis to identify and track profiles based on students' evaluations of scientific evidence and competing explanations, plausibility judgments of those explanations, and knowledge about science topics of social relevance. Three scientific thinking profiles emerged across time points—Pre-Scientific, Quasi-Scientific, and Scientific. The Pre-Scientific profile showed lower plausibility for scientific explanations but demonstrated the largest shifts in both plausibility judgments and knowledge gains. The Quasi-Scientific profile exhibited moderate knowledge scores and greater profile stability, whereas the Scientific profile reflected the highest knowledge and plausibility scores. At the first time point, 73.1% of students were classified as Pre-Scientific; by the final time point, 30.9% were Scientific, 39.8% Quasi-Scientific, and 29.3% Pre-Scientific. Beyond yielding these profiles, science learning gains were observed during the scaffolded lessons. This indicates that low-cost, well-designed scaffolds may be a promising avenue for differentiated instruction. The findings lay the groundwork for additional research and development of adaptive instructional strategies capable of developing scientific thinking across diverse student populations.

Educational Impact and Implications Statement

When learning about complex and sometimes controversial scientific topics of social relevance, middle and high school students do not all think alike—but many can shift toward a more scientific stance when supported with the right instructional tools. This study reveals three distinct profiles of scientific thinking and shows how students often move from less to more scientific patterns of understanding over time. These shifts occurred regardless of students' demographic or regional backgrounds suggesting that well-designed, evidence-based scaffolding can promote scientific thinking across diverse classrooms. By helping students critically evaluate competing explanations and reflect on which claims seem most plausible, teachers can promote deeper scientific understanding and civic engagement on scientific topics of social relevance.

Keywords: scientific thinking, educational scaffolding, person-centered analysis, plausibility judgments

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Researchers across fields and disciplines have sought to better understand the nature, characteristics, and development of scientific thought and knowledge construction. One goal has been to ensure that individuals are equipped not only to become agents of scientific and technological progress but also to engage with the consequences of scientific discovery in an informed and responsible manner (National Research Council, 2012). However, promoting students' scientific thinking is challenging due to the inherent complexity of many scientific topics. Scientific issues often intersect with socially relevant concerns (Owens et al., 2017). Scientific topics of social relevance involve concepts with significant ethical, economic, political, or environmental implications (e.g., the climate crisis). The spread of disinformation and science denial further complicates the situation (Aïmeur et al., 2023). Evidence suggests that consumption of and belief in disinformation interacts with higher-order thinking (Bronstein et al., 2019; Pennycook & Rand, 2020) and may unconsciously influence behavior (Bastick, 2021). Therefore, learning about scientific topics of social relevance, which involve competing and sometimes misleading claims, may especially require and challenge students' effective use of scientific thinking and reasoning (Presley et al., 2013; Sinatra & Lombardi, 2020).

Instructional scaffolding can support students' scientific thinking as they learn about scientific topics of social relevance. Empirical studies (for a summary, see Lombardi & Bailey, 2024) demonstrate that instructional scaffolding can help students make scientific evaluations when connecting evidence to competing explanations. Such scaffolding has been shown to shift students' plausibility judgments toward scientific conclusions and enhance their scientific knowledge. Other studies have shown that well-designed instructional scaffolding can be particularly effective in facilitating critical and evaluative thinking about scientific topics of social relevance (see, e.g., Darner, 2019; Dauer et al., 2022; Sadler et al., 2017). For example, science students may evaluate how well different lines of scientific evidence support competing explanations of phenomena. Bailey et al. (2022) suggested that evaluative processes may be particularly effective when learning about scientific issues of social relevance, which necessarily entail multiple, competing explanatory models. Lombardi and colleagues' (2016) theoretical framework posited that implicit and/or explicit cognitive evaluations are related to judgments about the truthfulness of claims (i.e., plausibility judgments), and students' learning may benefit when such judgments shift toward a more scientific stance.

Evidence for the broad effectiveness of instructional scaffolds in supporting scientific thinking and knowledge construction notwithstanding, this research has largely taken a variable-centered approach to examine relations between variables. Because variable-centered research may overlook meaningful patterns in students' scientific thinking, the present study adopted a person-centered approach to investigate distinct profiles that might inform differentiated instructional strategies. Although person-centered analysis is not uncommon in scientific thinking and epistemic cognition research (see, e.g., Kampa et al., 2016; Khan & Krell, 2021; Kuhn et al., 2000; Trevors et al., 2017), the present study served as a first to investigate profiles of scientific thinking characterized by the constructs of levels of scientific evaluation, plausibility judgments about competing explanations of scientific phenomena, and science knowledge, which are central to the development of scientific thinking (Lombardi et al., 2024).

This study builds on previous research examining students' evaluations of competing explanations of (socio-)scientific phenomena, alongside multiple indicators of their plausibility judgments and knowledge. It employs a longitudinal design to assess the stability of students' responses over the course of engaging in multiple scaffolded learning activities. Using contemporary latent profile and latent transition analyses, this research aimed to deepen the literature's understanding of scientific thinking and knowledge construction, while providing practical insights for teaching socially relevant scientific topics in classrooms.

Scientific Thinking

We situated the present study within the broader research tradition that views scientific thinking as an interactive, dynamic process integrating both cognitive and epistemic dimensions of reasoning and knowledge construction. Specifically, we drew upon the research synthesis presented in *A Framework for K12 Science Education* (National Research Council, 2012), which identified evaluative processes at the nexus of scientific practices. Philosophers and historians of science have long argued that evaluating competing explanations against evidence constitute a central task of science (Popper, 1959; Lakatos, 1970; Pickering, 1995). Complementing these normative accounts, developmental and educational psychologists document how learners come to coordinate theory and evidence in ways that mirror disciplinary expectations (Kuhn & Pearsall 2000; Lombardi et al., 2016). Furthermore, scientific argumentation and critique are key evaluative processes that support epistemic practices across all scientific disciplines (Osborne, 2010). Importantly, epistemic thinking specifically addresses how individuals conceptualize the nature, sources, and justification of knowledge. Contemporary accounts of epistemic cognition treat justification for knowing as central—namely, how individuals evaluate claims against evidence, weigh authority, and apply epistemic aims and values when adjudicating competing explanations (Barzilai & Chinn, 2024; Greene et al., 2008; Sinatra et al., 2014).

In contrast, scientific thinking encompasses additional cognitive and social processes involved in hypothesis generation, data collection, evidence evaluation, and explanatory model construction. These broader cognitive and social skills enable learners and scientists to navigate complex problems and phenomena beyond merely conceptualizing knowledge and knowing (Kuhn & Pearsall, 2000; Osborne, 2010). Within authentic scientific activities, such evaluations serve as a crucial bridge linking empirical evidence to theoretical explanations, ensuring explanations remain grounded in credible evidence and scientifically valid principles (Kuhn et al., 2017; NRC, 2012). Plausibility judgments represent a distinct epistemic outcome of scientific evaluation, reflecting individuals' assessments of the truthfulness of competing explanations (Lombardi et al., 2016). The evaluative process of scientific argumentation involves analyzing the logical structure, coherence, and evidentiary support of claims; plausibility judgments serve as the conclusion of such evaluations, shaping individuals' scientific thinking (McGrew, 2025; Nussbaum, 2011).

Lombardi and colleagues (2024) conceptualized scientific thinking as extending beyond fact acquisition to include a broad

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spectrum of cognitive and social processes. They saw scientific thinking as a dynamic, interactive process where individuals engage with scientific concepts and phenomena, often mediated by social interactions within a community of practice. Central to this is cognitive and social engagement with and critical evaluation of scientific claims in light of compelling alternatives (Sinatra & Lombardi, 2020). These evaluations may be shaped by factors such as cognitive biases, prior knowledge, and cultural contexts. Thus, scientific thinking involves engaging in scientific practices (e.g., asking critical questions, analyzing and interpreting data and evidence, and evaluating models and explanations) during the construction of scientific knowledge.

Science Knowledge

The construction (and reconstruction) of knowledge is an important aspect of scientific thinking. For example, knowledge that is consistent with the scientific consensus is often needed to ask meaningful questions, identify, analyze, and evaluate evidence, and generate scientific explanations (Lombardi & Bailey, 2024). Prior knowledge is a well-established determinant of successful learning across topics and domains (see Bittermann et al., 2023; for a recent comprehensive review). When prior knowledge is accurate and available, it serves as a foundation upon which new knowledge can be constructed. However, as Brod (2021) notes, prior knowledge can hinder learning if it conflicts with new information. The presence and prevalence of incomplete, unstable, or inaccurate prior knowledge may explain why it is not as reliable a predictor of learning as many assume (Simonsmeier et al., 2022). Many researchers consider prior knowledge a learning resource, even when inaccurate, as it can still serve as a scaffold for learning (see, e.g., Chen & Techawitthayachinda, 2021). In this way, instructional scaffolding may retool inaccurate prior knowledge to be used as a resource in the learning process (Kendeou et al., 2019; Lombardi & Danielson, 2022). Such scaffolding may facilitate students' cognitive organization of complex scientific topics and issues, help students to think and reason deeply to evaluate scientific evidence, and appraise the plausibility of multiple, often competing, claims (Lombardi et al., 2013). Therefore, any effective empirical investigation of scientific thinking must take scientific knowledge—its contents, construction, and dynamic connectivity to scientific practices—into account.

Scientific Evaluation

Scientific evaluation serves as a fundamental and iterative process in both scientific and science classroom communities. Within the scientific community, scientists and engineers systematically reflect on and assess their work, from the fit of models to observed data to the congruence of research design with theoretical frameworks (Schiefer et al., 2022). In the science classroom, student evaluation should emulate the authentic practices of scientists (Manz et al., 2020; NRC, 2012; Schriebl et al., 2023). In both scientific and science education contexts, critical evaluation involves assessing how evidence supports competing explanations of phenomena (e.g., theories, hypotheses, and models). For example, evaluation may require individuals to assess both the strengths and weaknesses of connections between evidence and explanatory models. More than mere critique, evaluations should address how effectively evidence supports not only a given explanation but also its plausible alternatives, such as counterarguments or contrary hypotheses (see, e.g., Cadillo-Benalcazar et al., 2021; Keane et al., 2021; Nauta et al., 2023). This approach aligns with the concept of falsifiability, which Popper (1959) posits as the discriminating principle between scientific knowledge construction and belief based on mythologies, religions, and superstitions. Falsifiability renders scientific explanations provisional, where reliable and trustworthy evidence has the potential to discredit one explanation in favor of an alternative. Although individual scientists may not always adhere strictly to the principle of falsifiability, the scientific community eventually discards hypotheses and theories that fail to hold up to evidence (Lakatos, 1970). This underscores the importance of critical evaluation in the scientific endeavor, as it helps refine and advance our understanding by constantly scrutinizing evidence and explanation and the dynamic relationship between the two.

Evaluations should extend beyond describing evidence concerning a single explanation. Instead, it should critically appraise connections between evidence and competing explanations, especially when a plausibility gap exists between scientific and lay perspectives (Lombardi et al., 2016; Sinatra & Lombardi, 2020). Here, *critical* evaluation refers to judgments about the relationship between evidence and alternative explanations of a particular phenomenon (Gans et al., 2024; Klavon et al., 2023). More critical evaluations may help students make more science-guided judgments when evaluating evidence and explanations (Ford, 2015). However, generating such evaluations may be especially difficult when students learn about scientific topics of social relevance. Seldom will a student enter the science classroom without any knowledge on a topic as pervasive as, say, the causes of climate change. Although such prior knowledge may often be inconsistent with scientific understanding of the current climate crisis, this knowledge may be useful in classroom discourse, where students explicitly evaluate connections between scientific evidence and competing explanations and work toward shifting their plausibility judgments toward a more scientific stance (Lombardi et al., 2022)

Plausibility Judgments

The development of scientific understanding and knowledge construction is a complex process that necessitates engaging with established scientific explanations (e.g., theories, hypotheses, claims, processes) and their underpinning evidence. Plausibility judgments may be central to the construction of scientific knowledge and result from implicit and/or explicit cognitive evaluations about the potential truthfulness of a given claim or explanation (Allen & Pardo, 2019; Lombardi et al., 2016). This construct plays a pivotal role in shaping reasoning about competing claims and the co-construction of knowledge in scientific communities, and it may be particularly useful when learning about scientific topics of social relevance (Sinatra & Lombardi, 2020).

According to Lombardi and colleagues' (2013, 2016) theoretical model, plausibility judgments are inherently tentative and often made implicitly through informal thinking processes with minimal reflective thought. Plausibility judgments often serve as the mental yardsticks by which individuals gauge the truthfulness of scientific explanations, often weighing them against alternative viewpoints based on their personal beliefs and prior experiences (Kahneman & Miller, 1986; Mercier, 2017; Posner et al., 1982). These

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judgments are not static; rather, they are easily changeable and open to revision and reappraisal in light of new information or evidence, which may make plausibility a dynamic driver of knowledge construction and reconstruction in certain classroom situations (Lombardi et al., 2016; Lombardi & Bailey, 2024). Plausibility judgments may become particularly salient when students engage with scientific topics of social relevance. In such situations, students may have to navigate the complexities of analyzing evidence, evaluating competing claims and explanations, and confronting plausibility gaps (Sinatra & Lombardi, 2020). However, scaffolding in science instruction can bridge this gap by encouraging students to explicitly evaluate the connections between scientific evidence and claims, thereby fostering shifts in their plausibility judgments toward a more scientific stance.

Although plausibility judgments often emerge from the same evaluative processes that underpin argumentation, they serve a distinct epistemic function (McGrew, 2025; Nussbaum, 2011). Specifically, argument evaluation entails weighing the logical structure, coherence, and evidentiary support of competing explanations, often in a dialogic context. Plausibility judgments capture a more global, sometimes implicit, sense of whether a claim appears more or less truthful. Argument evaluation is a procedural step—examining the strength of evidence and reasoning—while plausibility judgments are a possible outcome of that scrutiny, reflecting one's current stance on a claim's veracity (Lombardi et al., 2016). Understanding this distinction is key, as scaffolded instruction can guide students in strengthening their argument evaluation skills—by probing the validity of evidence, identifying gaps in reasoning, and countering contradictory interpretations—which shifts plausibility judgments toward more scientifically grounded conclusions.

Since the development of Lombardi and colleagues' (2013, 2016) theoretical model, many empirical studies involving middle school, high school, and undergraduate students have consistently tested and used the model as an effective way to promote more explicit plausibility (re)appraisals and knowledge construction of fundamental science concepts (see, e.g., Bailey et al., 2022; Dobaria et al., 2022; Medrano et al., 2020, 2025; Schoute et al., 2024). These studies have been part of a larger project developing instructional activities to help students make more critical and scientific evaluations between lines of scientific evidence and alternative explanations about scientific topics of social relevance.

Development of Scientific Thinking

Particularly relevant to the present study is the increasing role of evaluations about connections between evidence and explanation in developmental transitions. Kuhn and Pearsall (2000) posit that scientific thinking matures as individuals gain greater evaluative control over the coordination of theory and evidence. In terms of epistemic thinking, one can transition from an *absolutist* (prereflective) stance to a *multiplist* (quasireflective) stance to *evaluativist* (reflective) stance about knowledge and knowing (King & Kitchener, 2004; Kuhn et al., 2000). Those subscribing to an absolutist stance may view scientific knowledge as the sum of objective facts, and that reality is directly knowable. Those who hold a more multiplist perspective may view scientific knowledge as capricious, based on opinion and interpretation, with equal value placed upon individual, subjective perspectives. Importantly, "[t]his awareness initially assumes such proportions, however, that it overpowers any objective standard that could serve as a basis for evaluating conflicting claims. Because claims are subjective opinions freely chosen by their holders and everyone has a right to their opinion, all opinions are equally right" (Kuhn, 2010a, p. 515). Similarly, for those who hold a more evaluativist perspective, knowledge is also constructed via different perspectives; however, based on criteria and processes concerning reliability, trustworthiness, expertise, and truthfulness, some explanations have—tentatively—greater validity than alternatives.

Importantly, epistemic thinking alone does not fully characterize scientific thinking. Epistemic thinking focuses on how individuals conceptualize the nature, sources, and justification of knowledge (Chinn et al., 2011; Greene & Bråten, 2016; Hofer & Pintrich, 1997; King & Kitchener, 2004). Although this epistemic dimension is central in science, it constitutes only one aspect of the broader umbrella of scientific thinking, also including skills such as hypothesis generation, data analysis, and measurement. Scientific thinking involves both epistemic cognition and engagement in disciplinary practices to construct scientific knowledge (Lombardi & Danielson, 2022). In the course of evaluating lines of scientific evidence and competing claims, a more multiplist stance may make it difficult to be critically reflective to the degree necessary for conceptual development and learning consistent with scientific understanding (Bråten et al., 2014; Taber, 2025). A student's prior knowledge may predispose them to agree with one explanation of a scientific phenomenon over an alternative, but passing judgment about truthfulness of either explanation may not be amenable to such a subjective perspective. Therefore, developing a more evaluativist stance when considering scientific topics of social relevance may be preferential for science learning and the development of scientific thinking (Osborne et al., 2013).

It is also important to emphasize that—just as cognitive development broadly does not proceed in some "immaculate transition" (Siegler & Munakata, 1993)—the development of scientific thinking is not a neat, linear progression. Microgenetic research has revealed that children often exhibit overlapping waves of understanding, where new strategies emerge and coexist with older ones, rather than replacing them outright in a single step (Siegler & Jenkins, 2014). Through repeated, fine-grained observations over short intervals, microgenetic studies capture these fluctuating patterns, moments of insight followed by apparent backsliding, thereby illuminating the gradual and sometimes uneven shifts in children's reasoning (Kuhn, 1995; Xiao & Kuhn, 2024). Applied to scientific thinking, a similar dynamic may be observed: individuals vacillate between less and more scientific perspectives, drawing inconsistently on scientific practices and knowledge (Vosniadou, 2019). This protracted coexistence of and competition between more and less scientific ways of thinking has been demonstrated in adolescents, adults, and even in expert scientists (Allaire-Duquette et al., 2021; Potvin & Cyr, 2017; Shtulman & Valcarcel, 2012).

Instructional Scaffolding to Support Science Learning

Educational scaffolding provides temporary academic support to students as they learn and develop new skills or understanding (Wood et al., 1976). Drawing on Vygotsky's Zone of Proximal Development (Vygotsky & Cole, 1978), Wood et al. (1976)

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described scaffolding as a process where a more knowledgeable other controls "those elements of the [learning] task that are initially beyond the learner's capacity, thus permitting him [*sic*] to concentrate upon and complete only those elements that are within his range of competence" (p. 90). Instructional scaffolding has been adapted across educational levels and subject areas, demonstrating its versatility and effectiveness in enhancing student learning (see, e.g., Lin et al., 2012; Puntambekar, 2022; Van de Pol et al., 2010). Contemporary research continues to explore the nuances of effective scaffolding. Studies focus on how different types of scaffolds—such as procedural, conceptual, metacognitive, and strategic scaffolds (Hannafin et al., 2013)—support various aspects of learning. Investigations of the relationship between more autonomy-supportive scaffolding have also emerged (see, e.g., Collie, 2022; Schoute et al., 2024).

Science learning in the classroom, and learning about scientific topics of social relevance in particular, presents a unique opportunity to leverage the advantages of educational scaffolds. The challenge for students to engage meaningfully with and think scientifically about scientific issues of social relevance is twofold: these issues are often complex, and they seem to attract competing, nonscientific explanations, even conspiracies (e.g., "climate change is a hoax;" Dawson & Eilam, 2022; Levrini et al., 2019; Peel et al., 2017; Sinatra & Hofer, 2021). Educational scaffolds can be facilitative in both respects. Bailey et al. (2022) reported that particular instructional scaffolds, called Model-Evidence Link (MEL) activities, which were designed to facilitate students' scientific evaluations of the connections between lines of scientific evidence and competing explanations about the providence and effect of the climate crisis, deepened high school and undergraduate college students' learning. These researchers built upon the notion that making more scientific evaluations can support students' critical thinking (Ford, 2015). Lombardi and colleagues (2013, Lombardi, Bickel, et al., 2018, Lombardi et al., 2022) specifically designed a series of MEL scaffolds to facilitate explicit evaluation between scientific evidence and competing claims, with classroom-based research revealing statistically significant and educationally meaningful shifts in students' plausibility judgments toward the scientific consensus and deeper understanding of science concepts when using the MEL. This work (for a summary, see Lombardi & Bailey, 2024) continued a long line of research examining the effectiveness of MELs versus other types of instructional scaffolds that teachers had used in their science instruction. The sum of this research suggests that MEL diagrams can (a) facilitate students' evaluations of connections between lines of scientific evidence and competing explanatory models, (b) shift students' plausibility judgments toward a more scientific stance, and (c) enhance students' knowledge of core scientific ideas and practices.

Scaffolding can reinforce scientific thinking, preventing students from reverting to intuitive or less scientific conceptions. Research suggests that scaffolds promoting explicit evidence evaluation, model-based reasoning, and metacognitive reflection are especially effective at solidifying conceptual change (e.g., Kelemen et al., 2013; Potvin & Cyr, 2017). For instance, repeatedly guiding learners to justify their claims with data, consider alternative explanations, and critically compare empirical results to theoretical models helps them internalize scientific practices rather than rely on "common sense" assumptions. Moreover, scaffolding that fades gradually—giving students increasing autonomy for carrying out investigative and argumentative tasks—encourages them to monitor their reasoning and maintain a rigorous, evidence-based mindset long after the support is withdrawn (McNeill et al., 2006). Consequently, scaffolded science lessons not only teach facts and concepts but may also foster enduring habits that are resilient to the everyday pull of intuitive misconceptions. However, although prior studies using MEL scaffolds or other similar approaches have demonstrated important gains through variable-centered analyses, they may obscure meaningful heterogeneity in how individual students develop scientific thinking. To address this, the present study adopted a person-centered approach to identify and examine distinct profiles of scientific thinking and how they evolved.

The Present Study

This study is part of a 6-year design-based research project on developing and implementing MEL scaffolds in middle and high school science classrooms. Design-based research projects typically examine instructional effectiveness through an iterative process (Barab & Squire, 2016). The research team—including classroom teachers, science education researchers, educational psychologists, and scientists—created all materials (e.g., lines of evidence and alternative explanations included in the scaffolding) through a design-based process, involving iterations of development, testing, analysis, and review. The team used primary source documents from highly ranked scientific journals (e.g., *Science*, *Nature*) in developing these materials. After multiple iterations of development and testing, an independent panel of scientific and pedagogical advisors reviewed the materials for accuracy, precision, reliability, and face and content validity. For the present study, we used data collected in classrooms during the project's fifth year (the 2021–2022 academic school year), which was the final year of data collection.

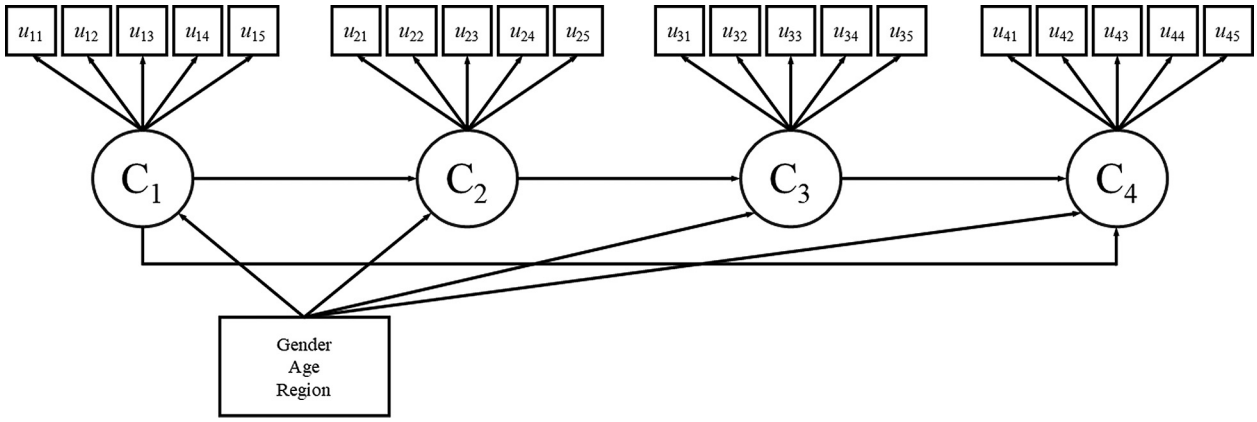
Research Questions

Our primary research question asks, are there distinct profiles of scientific thinking evident in adolescent students engaged in science learning and, if so, how many of these profiles exist? We also sought to determine the extent to which students may change or transition into new profiles over time. Last, we examined the extent to which age, gender, and geographical region predicted initial profile membership or subsequent transitions between profiles over time. The conceptual model guiding our research questions is illustrated in Figure 1.

We approached this study with the a priori assumption that our sample is heterogeneous, with no explicit hypothesis regarding the exact number of profiles that will emerge from the sample (Masyn, 2013). Given the nature of mixture models, we were prepared for an outcome where numerous, statistically significant multiprofile solutions were enumerated (Nylund et al., 2007). Given our previous research on the constructs of interest, we predicted that profiles characterized by higher knowledge scores would also be characterized by plausibility judgments more aligned with scientific consensus (see Lombardi & Bailey, 2024). We also predicted that profiles for which there is a large positive "shift" in plausibility ratings pre- to postinstruction will be characterized by larger knowledge gains as well.

Grounded in Kuhn's (2010b) epistemic cognition framework, theories of conceptual change (Posner et al., 1982), and evidence suggesting that scaffolds supporting reflective thinking facilitate deeper inquiry (Reiser, 2018; White & Frederiksen, 1998), we

Figure 1
Latent Transition Model for Indicators of Scientific Thinking Over Four Time Points



Note. This figure represents a latent transition model diagram with five observed variables for each instructional episode (evaluation, measurements of plausibility at pre- and postinstruction, and measurements of knowledge at pre- and postinstruction) at four episode time points (C_1, C_2, C_3, C_4), with first- and second-order effects. Gender, age, and region are represented as covariates. Errors not included for model clarity.

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hypothesize that—across multiple scaffolded lessons—students will transition to increasingly sophisticated scientific thinking profiles (i.e., higher levels of science knowledge, higher levels of evaluation, and plausibility judgments more aligned with the scientific consensus). However, empirical work indicates that not all students follow a purely linear, less-to-more progression in the development of their scientific thinking and reasoning, neither across a single instructional intervention (e.g., Ferguson & Bråten, 2013) nor across multiple years of instruction (e.g., Lazonder et al., 2021). Consequently, the overarching prediction is that a meaningful proportion of students will transition from profiles characterized by less scientific thinking to profiles characterized by more scientific thinking, but with the understanding that dynamism—including plateaus and even regression—will be present within this greater trend.

Our prior research has shown that educational scaffolds can successfully facilitate science learning in middle school, high school, and undergraduate classrooms. However, given the role of prior knowledge in learning, along with the general rule that older students tend to have more experience in the classroom and with more varied academic topics than younger students, we predicted that age would have utility in predicting profile membership and transitions (Bittermann et al., 2023). Although scientific thinking may not naturally develop, adolescence is attended by the continued development of basic and higher-order cognitive functions that could support scientific thinking (Kuhn, 2006). These developments are neither singular, uni-directional, nor linear, but do proceed as a function of time. Including student age in our analysis would also contribute to the "need to conduct more studies of adolescent cognition in the situated contexts of the activities in which teens choose to invest their intellectual resources" (Kuhn & Franklin, 2008, p. 544).

Participants in this study were drawn from two distinct geographic regions: the United States Southeast and Mid-Atlantic, each with unique educational landscapes and sociocultural contexts (Ebert et al., 2022; Rentfrow et al., 2013). Given documented regional variations in science curricula (Jin et al., 2019) and potential differences in exposure to scientific topics of social relevance, it is possible that latent differences in scientific thinking may partially reflect these contextual influences. However, considerable heterogeneity exists within each region, including variations in socioeconomic status, school resources, and individual student backgrounds. Our focus on regional grouping provides a broad starting point for understanding patterns potentially linked to differing educational practices and opportunities.

The present study aimed to explore the degree to which student engagement in the MEL activities elucidates distinct profiles of scientific thinking. In addition, we sought to ascertain whether these profiles maintain stability throughout an academic year, following a student's engagement with multiple MEL activities. Using a person-centered approach, we can identify whether or not likely groupings of students with similar ways of reasoning about scientific phenomena exist within our sample (Bergman & Trost, 2006). To our knowledge, no other study has attempted to define profiles of scientific thinking that incorporate combinations of students' knowledge, plausibility judgments, or their evaluations of science topics. This further presents a novel opportunity to characterize students' scientific thinking and how their understanding of scientific phenomena may change over time after repeated exposure to scaffolded activities.

Method

Participants

Middle and high school students taught in six different Earth and environmental science classrooms participated in the study. Teachers provided informed parental consent and student assent forms, with 366 students providing assent and consent. We recruited participants from the Mid-Atlantic (n = 77) and Southeast (n = 289) regions of the United States. Among them, 51.6% (n = 188) self-reported as White, 25.6%; (n = 94) as Asian, 11.1%; (n = 40) as Hispanic or Latino, 5.76%; (n = 21) as Black or African American, 5.67%; (n = 20) as two or more races,

5.46%; and less than 1% (n = 3) as American Indian or Alaskan Native. In addition, our sample included students in middle school, grade 6 (n = 235, 64.2%; M_{age} = 11.1 years, SD =. 343) and high school, grades 9 through 12 (n = 131, 35.8%; M_{age} = 15.4 years, SD = 1.51).

Research Design

The present study used a counterbalanced, quasiexperimental design, where participants engaged with four MEL activities over an academic year. Each participant experienced both forms of the MEL scaffold (pcMEL and baMEL, details below), but all completed the pcMEL first and the baMEL second. The third and fourth MELs were counterbalanced, with half the participants experiencing the pcMEL third and baMEL fourth, and the other half experiencing the baMEL third and pcMEL fourth. The sequence of the pcMEL first follows the historical approach of introducing scaffolded instruction, where students learn both concepts (e.g., topic knowledge about scientific topics of social relevance) and skills of inquiry (e.g., evaluating evidence to explanation connections, in light of alternatives; Bruner et al., 1956; Posner & Strike, 1976). In the present study, we did not include a business-as-usual control group because previous, variable-centered studies comparing MEL scaffolding with "typical classroom science instruction" revealed appreciable and meaningful advantages in student performance with the MELs (see, e.g., Lombardi et al., 2013, Lombardi, Bailey, et al., 2018). Therefore, given the demonstrated advantages of MEL scaffolding over traditional science instruction, we determined that including a control group would be unwarranted and possibly unethical (see, e.g., British Educational Research Association, 2024; Joyce & Cartwright, 2018; Troyer, 2022).

During the implementation of the MEL activities, we collected data regarding students' thinking about scientific topics of social relevance on the following topics: the climate crisis (n = 197), fossils and fossil fuel use (n = 288); water resources (n = 316); and cosmic origins (n = 326). This resulted in a total of 1,191 observations. The order and sequence of these topics varied by school based on the curricular scope and sequence. To account for this variation, we made some transformations to the data collected to arrive at a fair comparison among classes (detailed in the later section), which still allows for considerations of ecological validity that more closely reflect classroom learning environments.

All intervention activities were delivered by participants' regular classroom teachers. In preparation for the study, participating teachers attended 2–3 day, in-person or virtual workshops that introduced the theoretical underpinnings of the MEL (e.g., model-based reasoning, plausibility judgments, evidence evaluation); step-by-step guidance on facilitating lessons and how to support student reading of evidence texts; and sample lesson videos and "teach-back" exercises, in which teachers practiced presenting a MEL activity to peers. Where possible, local members of the research team observed or recorded 1–2 MEL lessons per teacher to confirm adherence to the recommended procedures. Observations were semistructured: An observer used a standard protocol to note teacher prompts, time allocated to each step, and overall student engagement. Generally, teachers were encouraged to act as facilitators rather than lecturers during MEL activities. They introduced the topic, directed students to read evidence texts, clarified diagram instructions, and prompted reflection. Teachers typically provided procedural feedback (e.g., how to annotate the diagram, how to interpret evidence statements) but not correctness feedback on plausibility ratings or model-evidence linkages. Some teachers held short whole-class discussions afterward to highlight interesting or strong student justifications.

Before participants engaged in any of the scaffolds, they participated in a short learning task about how scientists connect lines of evidence and explanatory models, first defining plausibility as "a judgment we make about the potential truthfulness of one model compared with another and the judgment may be tentative (not certain)." Both the pcMEL and baMEL followed a similar instructional progression. First, participants completed a preactivity questionnaire to gauge topic knowledge. Second, they rated the plausibility of the scientific and alternative explanatory model(s); models are labeled with letters *A*, *B*, and where applicable, *C*, but are not identified as *scientific* or *alternative*. Third, participants read one-page expository texts for each line of evidence (i.e., four expository texts for pcMEL, eight expository texts for baMEL). Fourth, participants completed their MEL diagrams. Fifth, participants gave plausibility ratings for the same explanatory models as before the intervention and then wrote explanations for two of their generated model-evidence links. Sixth, participants completed a postinstruction knowledge questionnaire identical to the preinstruction measure (see the supplemental materials, e.g., activity worksheets for both pcMEL and baMEL). The entire time of each MEL lesson was approximately 90 min (about two traditional class meetings or one block class meeting).

MEL Scaffolds

The pcMEL scaffold provided participants with a preconstructed diagram featuring two alternative explanatory models explaining a scientific phenomenon of social relevance, with one model being the scientific consensus model and the other being a plausible but nonscientific alternative (see Figure 2). For example, in the causes of climate change pcMEL, Model A declares that current climate change is caused by human activities (the scientific consensus explanation), whereas Model B states that current climate change is caused by an increasing amount of energy received by the Sun (a nonscientific but plausible explanation). The activity presents these two explanatory alternatives without any indication of the veracity or validity of either. The pcMEL also presented four lines of scientific evidence in the form of one-to-three-sentence declarative statements. One-page evidence texts accompanied each line of evidence to provide students more detail.

Participants constructed their diagrams in the baMEL activity by selecting two explanatory models from three competing alternatives. Students were not made explicitly aware of which of the alternatives represented the scientific consensus. Participants also selected four lines of scientific evidence from eight options. Compared with the pcMEL, this degree of choice was intended to elicit more autonomy and thereby engagement on the part of participants (Bailey et al., 2022; Schoute et al., 2024).

Measures

All data-collection procedures were fully disclosed to and approved by university Institutional Review Board and each

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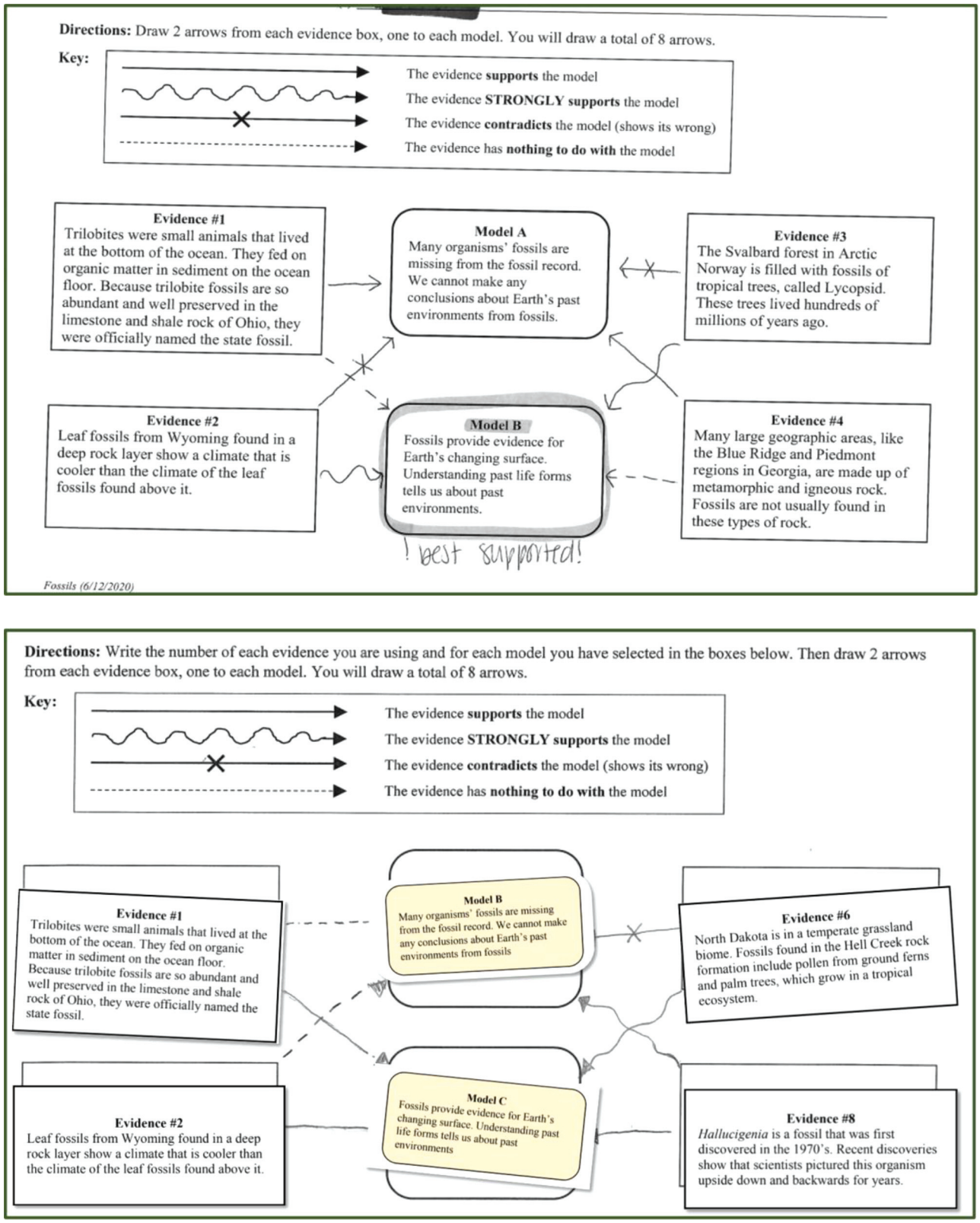
Figure 2

Examples of Model-Evidence Link (MEL) Diagrams



Note. Preconstructed MEL (pcMEL; top), build-a-MEL (baMEL; bottom). See the online article for the color version of this figure.

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participating school district's research office. Parental consent forms explicitly stated that all classroom work products generated during normal instruction could be analyzed for research purposes. Students were made aware of the same and provided written assent. During lessons, data collection was embedded in the learning process to mitigate performance anxiety and enhance ecological validity.

Knowledge

Before and after each MEL activity, students completed a short knowledge questionnaire on that MEL's topic. The questionnaires covered the topics of climate change (16 items), fossils and fossil fuel use (16 items), water resources (16 items), and cosmic origins (19 items). Students completed the same questionnaire at pre- and postinstruction. Based on the methods used in earlier MEL studies (see, e.g., Lombardi et al., 2013, Lombardi, Bailey, et al., 2018), students rated each knowledge item on a 5-point Likert scale (1 = *strongly disagree* to 5 = *strongly agree*), indicating the extent to which they believed scientists would agree with the statement. Having students rate their level of agreement from a scientist's point of view reflected their knowledge of scientific phenomena rather than their personal beliefs or opinions on the topic (Lombardi et al., 2013). The MEL project team developed these statements from information on which there is clear scientific consensus, with at least one question addressing each evidence statement and at least one question addressing each explanatory model.

We assessed the internal consistency of our knowledge questionnaires using Bayesian unidimensional Guttman λ_2, which offers advantages over traditional reliability measures like Cronbach's α. Unlike α, which assumes τ-equivalence and often underestimates reliability when item variances differ, λ_2 provides a more flexible and accurate measure by incorporating item covariances (Guttman, 1945; Pfadt et al., 2023). In addition, some have similarly proposed using other alternative metrics such as McDonald's ω and the greatest lower bound (GLB), but each has limitations; ω depends on the assumption that a unidimensional factor model fits the data well, whereas GLB tends to overestimate reliability unless the sample size is very large (Sijtsma, 2009; Ten Berge & Socan, 2004). Given these trade-offs, λ_2 was particularly well-suited for the present study, as it provided a robust lower bound for reliability without the restrictive assumptions of α or the potential inflation seen in GLB (Oosterwijk et al., 2016). Furthermore, the Bayesian approach enhances our reliability estimation by incorporating prior information, providing credible intervals that account for uncertainty, and performing robustly even when estimating reliability across different topics (Pfadt et al., 2023). By using Bayesian λ_2, we ensured that internal consistency estimates would not be misinterpreted as fixed values but rather as distributions that reflect the inherent uncertainty of our knowledge measurement.

Our estimated reliability indices were $\lambda_2 = .659$ for climate change knowledge items; $\lambda_2 = .670$ for water resources knowledge items; $\lambda_2 = .490$ for origins knowledge items; and $\lambda_2 = .765$ for fossils and fossil fuel. Although three of the four topics fell below the .7 heuristic of acceptability—often attributed to Nunnally (1978)—later work by Nunnally and Bernstein (1994) emphasized that values lower than .7 may be acceptable in certain research situations (e.g., conducting classroom-based research where students are offered a relatively high level of autonomy and choice, as was the case with the MEL scaffolding). For example, Flake et al. (2017) reported that measures can exhibit lower reliability, particularly in learning and reasoning research. McNeish (2018) and Taber (2018) also argued that reliability should be evaluated in the context of a measure's purpose, rather than adhering to a heuristic-based threshold. We designed our knowledge questionnaires for more granular level measurement than would be achieved by, for example, more conventional multiple-choice items each with a single correct answer. This may naturally result in lower internal consistency due to intentional item heterogeneity. Such heterogeneity is essential for capturing the complexity of the construct and does not necessarily indicate an unreliable measurement (Wallace & Bailey, 2010). Reliability must also be considered alongside validity and theoretical appropriateness (Sijtsma, 2009). In classroom-based educational research, measures with lower reliability may still provide valuable insights, particularly when they capture meaningful individual differences and are supported by strong theoretical grounding. Given that our previous studies have found similar reliability patterns (e.g., Lombardi, Bailey, et al., 2018, Lombardi, Bickel, et al., 2018), our present findings suggest that reliability values below .7 can still be informative. Thus, we concluded that our knowledge questionnaires provided meaningful and valid insights for the present study, despite the conventional yet sometimes overstated acceptability heuristic of .7

To account for differences in the four topic themes (climate change, fossils and fossil fuel use, water resources, and cosmic origins), we linearly transformed knowledge scores using min–max normalization (Jain et al., 2005). This approach standardizes each topic's score distribution to a common 0–1 scale, where the minimum observed value becomes 0 and the maximum becomes one. This approach is commonly used in medical, computing, and environmental research (Mazziotta & Pareto, 2022). Using such a method maintains the original distribution of the data as long as there are no outliers in the sampled scores (Mu, 2020). Therefore, before employing this technique, we screened for any score outliers and found none.

Levels of Evaluation

Participants constructed their written responses after completing their MEL diagrams. Following a prompt, participants indicated the MEL they wanted to discuss including its strength and type of relation (i.e., strongly supports, etc.). Participants then generated written explanations for why they made the given link. To gauge participants' level of evaluation, Lombardi and colleagues (2016) developed a rubric for scoring written responses (see the supplemental materials for rubric code definitions and characteristic student evaluations). Two independent scorers read each written explanation and used the rubric to score participants' levels of evaluation, from 1 (*an erroneous evaluation*) and 2 (*a descriptive evaluation*) to 3 (*a relational evaluation*) and 4 (*a critical evaluation*). A one-way random effects model for intraclass correlation coefficient was calculated to be .85, an acceptable level of scorer reliability. After scoring explanations, the scorers met to discuss any scoring differences and reached a full consensus on all scores. We used these consensus-based scores for subsequent analyses.

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Plausibility Judgments

Participants rated the plausibility of each explanatory model on a scale from 1 (*highly implausible*) to 10 (*highly plausible*), both before and after they completed each MEL activity. Calculations for a single plausibility judgment score differed for the two types of instructional scaffold. On preconstructed MELs, participants rated the plausibility of one scientific explanation and one alternative explanation. We calculated the plausibility judgment score by subtracting the alternative model's plausibility rating from the scientific model's plausibility rating. On build-a-MELs, participants rated one scientific explanation and two alternative explanations. We first averaged the plausibility ratings of the two alternative models and then subtracted this average from the scientific model's plausibility rating:

The resulting plausibility judgment scores range from −9 to +9. A positive score indicates that the participant judged the scientific explanation as more plausible than the alternative(s) (i.e., a more scientific stance) while a negative score indicates a stronger belief in one or more alternative explanations (i.e., a less scientific stance). A score of 0 denotes equal plausibility ratings for the scientific and alternative models.

Analytical Methods

We conducted a series of analyses to examine patterns in students' scientific thinking over time. First, we assessed and addressed missing data. Next, we performed latent profile analyses at each instructional time point to identify distinct profiles of scientific thinking based on students' levels of evaluation, plausibility judgment, and knowledge scores. After determining the optimal number of profiles, we extended the analysis by conducting longitudinal latent profile analyses to assess profile stability and changes over time. Finally, we employed latent transition analysis to model within-person transitions between profiles and examined demographic predictors of profile membership and transitions. This analytical sequence of steps is similar to other psychological and educational research studies (see, e.g., Fryer, 2017; Gillet et al., 2017; Schlatter et al., 2021). The following subsections detail each of these steps in the analytical process.

Missing Data

From our initial data set, we identified that over 30% of items—collected from each student across each of the four time points—were either entirely missing or uninterpretable (e.g., a student selected multiple choices for a single item). Furthermore, for individuals who had half or more of their data missing, essentially only completing two or fewer MEL activities, we could not deduce a pattern that informed or justified our assumption the data were missing at random. Therefore, we chose to exclude participants who were missing data equivalent to two or more complete time points (>20% missing). The decision to include students with up to 20% missing data is consistent with best practices in handling missing data in school-based studies where attrition rates of 15%–20% are common (Peng et al., 2007). This left us with a final sample of 197 students. The remaining missingness within our sample is displayed in Table 1.

Our final sample had less than 10% missing data, which we deemed acceptable for subsequent analyses. Within the 197 students who met this criterion, 34 fully completed all four MELs, 162 completed four MELs with missingness, and only one student completed three MELs. The average therefore being 3.99 MELs completed per student within this sample. We used full information maximum likelihood (FIML) in Mplus Version 8 (Muthén & Muthén, 1998-2017) to estimate missing values. FIML is preferable to multiple imputation, which can introduce bias if the imputation model is misspecified or if the missing data mechanism is not understood (Enders, 2010; Graham, 2009; Hayes & Enders, 2023).

Latent Profile Analyses

We first conducted a series of latent profile analyses (LPA) that were estimated separately at each time point to deduce the number of profiles that could be consistently extracted across time. For each time point, we initially set out to extract 1–10 profile solutions, in which the means and variances for each measure of evaluation, preplausibility, postplausibility, preknowledge, and postknowledge were estimated freely in all profiles (Peugh & Fan, 2013). When examining profile solutions for Time 1, our models ceased to converge when modeling seven or more profile solutions. Therefore, we only estimated 1–6 profile solutions for the remaining time points. To avoid local maximum, we formed all LPA using 5,000 random sets of starting values and 500 iterations, retaining the 250 best final solutions for optimization. Results for these analyses can be found in the LPA estimation file, Table S1 within our supplemental materials.

To determine the optimal number of profiles across timepoints, we evaluated several criteria including the meaningfulness of profile solutions, their theoretical justification, as well as findings produced from several statistical indices. The fit indices used included the Akaike information criterion (AIC), constant AIC (CAIC),

Table 1
Percentages of Missing Data Across MELs Collected at Each Time Point

	Time 1			Time 2			Time 3			Time 4		
Measures	*N* missing	Total items	%	*N* missing	Total items	%	*N* missing	Total items	%	*N* missing	Total items	%
Evaluation	68	394	17.3	89	394	22.6	53	394	13.5	33	394	8.38
Preplausibility	49	394	12.4	28	591	4.74	19	502	3.78	22	476	4.62
Postplausibility	28	394	7.11	122	591	20.6	43	502	8.57	34	476	7.14
Preknowledge	72	1,312	5.49	67	2,198	3.05	72	2,311	3.12	101	2,167	4.66
Postknowledge	510	1,312	38.9	109	2,198	4.96	107	2,311	4.63	68	2,167	3.14

Note. These data represent the final sample size (n = 197). Total number of items with missing data across all four time points is 1,694, resulting in 7.89% missingness overall.

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Bayesian information criterion (BIC), and sample-size adjusted BIC (ABIC), where lower values indicate better model fit. We also considered entropy values to assess classification accuracy, with values closer to one indicating better separation between profiles. In addition, we employed likelihood ratio tests—the Lo-Mendell-Rubin adjusted likelihood ratio test (ALMR) and the bootstrap likelihood ratio test (BLRT)—where a significant p value suggests that the model with k profiles fits better than the model with one fewer profile. Each of these indices is also reported in Table S1 for transparency and to allow for future comparisons of scientific thinking profiles that incorporate MEL materials.

Longitudinal LPA

Once we identified the most appropriate number of profiles for each time point, we combined the four separate LPA solutions into a single longitudinal model to assess the similarity of profiles over time. This sequential strategy follows guidance for best practices outlined by Morin et al. (2016). Findings from these analyses can be found in Table 2. The first step in this sequential strategy involved testing for configural similarity, which assesses if the same number of profiles can be extracted at each time point in alignment with the previous time-specific LPA findings. If this is achieved, then we proceed with incorporating additional constraints. The second step is evaluating structural similarity, where equality constraints are only imposed on the means for each of the profile indicators across time points. If the structural similarity holds, we then examine dispersion similarity as the next step in this process. Here, the means retain their constraints from before, and we additionally impose equality constraints on their variances as well. The final step in this process assesses distributional similarity, where equality constraints are imposed on each class probability across the four time points. The fit of these models is evaluated similarly to the aforementioned criteria.

Latent Transition Analyses

The model with the best fit from the former sequential strategy is retained and integrated within a longitudinal latent transition analysis (LTA) model. Within this model, we further evaluated within-person stability and the transitions observed in profile membership. Subsequently, we progressed with the most similar retained LTA model to investigate the predictive similarity of the profiles. This process necessitated the inclusion of several demographic variables (including gender, age, and region) within our model to explore potential associations between these predictors and profile membership. We further considered the effect of MEL order (i.e., whether the first, second, third, or fourth MEL activity) on profile membership to determine if topic effects lead to differences in profile membership. Logistic regressions were used to evaluate these associations and determine the likelihood of profile membership. When evaluating the performance of these predictors, we considered three models.

The first model examined the relationships between the predictors and profile membership, estimating these relationships freely across time points, while also permitting the predictors at Time 2 to vary in relation to Time 1. The second model allowed for the free estimation of the predictors' effects across time, without variation within profiles. Finally, we assessed the overall predictive similarity of the model by constraining the predictors to be equivalent across time. The fit of each model was evaluated, and the best-fitting model was selected for further interpretation. The resulting regression coefficients elucidate the effects of the predictors on the log odds of a potential profile membership outcome, contingent upon a one-unit increase in the predictor. To enhance the clarity of these results, we also present the odds ratios, which reflect the likelihood of a student's membership within a specific profile compared with another profile, contingent upon each unit increase in the predictor. These findings are presented in Table 3.

Transparency and Openness

Study materials (e.g., additional tables and figures, examples of instructional material, anonymized data, coding scheme, and software output/scripts) are available via the Open Science Framework: https://osf.io/7y4jd/files/osfstorage?view_only=394e03b50

Table 2
Results From Latent Profile and Latent Transition Analyses for the Three-Profile Solution

Model	LL	#fp	Scaling	AIC	CAIC	BIC	ABIC	Entropy
Step 1: Latent profile analyses estimated separately at each time point								
Time 1 (n = 195)	−1,253.819	22	1.023	2,551.639	2,645.645	2,623.645	2,553.952	0.704
Time 2 (n = 196)	−1,017.033	22	1.176	2,078.067	2,172.185	2,150.185	2,080.492	0.704
Time 3 (n = 196)	−1,103.574	22	1.162	2,251.149	2,345.267	2,323.267	2,253.573	0.712
Time 4 (n = 196)	−1,178.827	22	1.171	2,401.655	2,495.773	2,473.773	2,404.080	0.737
Step 2: Longitudinal latent profile analyses								
Configural similarity	−4,481.456	128	1.095	9,218.911	9,766.510	9,638.510	9,233.018	0.715
Structural similarity	−4,548.492	83	0.981	9,262.983	9,618.067	9,535.067	9,272.131	0.702
Dispersion similarity	−4,603.122	38	1.387	9,282.244	9,444.812	9,406.812	9,286.432	0.652
Distributional similarity	−4,658.175	32	1.673	9,380.349	9,517.249	9,485.249	9,383.876	0.620
Latent transition analysis	−4,577.165	54	1.238	9,262.330	9,493.348	9,439.348	9,268.281	0.702
Step 3: Predictive similarity								
Profile-specific free relations with predictors	−5,050.869	67	0.582	10,235.738	10,522.372	10,455.372	10,243.123	0.723
Free relations with predictors	−5,056.370	49	0.918	10,210.741	10,420.368	10,371.368	10,216.141	0.728
Equal relations with predictors	−5,093.259	31	0.932	10,248.519	10,381.140	10,350.140	10,251.936	0.670

Note. LL = model loglikelihood; #fp = number of free parameters; Scaling = scaling factor associated with the maximum likelihood estimates using robust standard errors; AIC = Akaike information criteria; CAIC = constant AIC; BIC = Bayesian information criteria; ABIC = sample-size adjusted BIC.

Table 3
Logistic Regression Coefficients and Odds Ratio With Demographics as Covariates Using Quasi-Scientific Profile as the Comparison Group

Latent profiles	Effect	Coefficient	*SE*	*Z*	*p*	Odds ratio
Pre-Scientific	Gender	−.070	.248	−0.282	.778	0.932
	Age	−.016	.031	−0.531	.595	0.984
	Region	.098	.206	0.479	.632	1.104
Scientific	Gender	−.122	.267	−0.458	.647	0.885
	Age	.053	.033	1.613	.107	1.055
	Region	−.374	.232	−1.610	.107	0.688

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4a3431292b59c5ab93a2030. Data were analyzed using Mplus Version 8. This empirical study was not preregistered.

Results

Latent Profile Solutions

The results for each profile solution estimated separately across the four time points are presented in the LPA estimation file, Table S1 in the supplemental materials. When examining these fit indices, we noticed that at each time point, indicators improved with the addition of profiles to the solutions, with the exception of the ALMR values, which suggested a two-profile solution at Time 1, Time 3, and Time 4. However, considering ALMR is known to tend to underextract the correct number of profiles (Gillet et al., 2017), we did not rely on these values exclusively. When evaluating the entropy of each solution, we found similar results across each profile solution (.640–.821) suggesting acceptable to high values. The BLRT shows consistent evidence for a three-profile solution across each time point and partial evidence for a four-profile solution for Times 2, 3, and 4.

To extract the correct number of profiles, we further examined several elbow plots (Morin et al., 2016), which are displayed in the LPA estimation file, Figures S1–S4 in the supplemental materials. These plots reveal the improvement in fit appears to flatten out between the three and five profile solutions. These plots also aided in confirming that moving from the two- to three-profile solution provides an additional meaningful and qualitatively distinct profile across all four time points. When considering the four-profile solution, results showed inconsistent separations of the prior profiles and provided mixed findings in improved fit for three of the fit indices (CAIC, BIC, BLRT). The three-profile solution was thus retained at each time point, and the final time-specific LPAs are reported in Table 2.

At four time points an LPA of configural similarity, with three profiles per time point, was then estimated. Next, we compared this model with a model of structural similarity, by constraining the within-profile means on the knowledge, plausibility, and evaluation indicators to be equal across time points. Compared with the model of configural similarity, this model resulted in lower values on the CAIC and BIC, thereby supporting the structural similarity of this solution across time points. We then estimated where the within-profile variance of each indicator was constrained across time points to assess the dispersion similarity of our model. Compared with the model of structural similarity, this model resulted in lower values for CAIC and BIC, supporting the dispersion similarity of the solution. Finally, we estimated a model of distributional similarity by constraining the size of the latent profiles to be equal across time points. When compared with the dispersion model of similarity, the distributional similarity model did not result in lower values on any of the information criteria. The model of dispersion similarity was therefore retained for subsequent analyses.

Three-Profile Solution

The indicator mean estimates for each of the three profiles in the model of dispersion are shown in Table 4 and Figure 3. Profile 1 is characterized by the lowest evaluation score (M = 1.56) among the three profiles, lowest pre- (M = −.64) and postinstruction (M = .69) plausibility scores, and the lowest pre- (M = 3.37) and postinstruction (M = 3.45) knowledge scores. Notably, this group was the only profile associated with a negative plausibility judgment estimate (at preinstruction), indicating a stance where the nonscientific alternative explanation is judged to be more plausible than the scientific explanation. This profile also had the largest pre-to-post shift in plausibility judgments. We chose to label this group the *Pre-Scientific* profile. Profile 2 is characterized by the highest evaluation scores (M = 1.82), the highest plausibility judgments both pre- (M = 4.97) and postinstruction (M = 5.50), and the highest knowledge scores at both pre- (M = 3.94) and postinstruction (M = 4.17). We labeled this the *Scientific* profile. Estimates for Profile 3 fell between those of the Scientific and Pre-Scientific groups, with evaluation score almost as high as the Scientific profile (M = 1.80), moderate plausibility judgments pre- (M = 2.85), and postinstruction (M = 3.11), and moderate knowledge pre- (M = 3.44) and postinstruction (M = 3.57). We labeled this group the *Quasi-Scientific* profile.

Table 5 and Figure 4 present the proportion of students in each profile across the four measurement points, revealing shifts in how many learners occupy each profile over time. At Time 1, the largest proportion of students was classified as Pre-Scientific (73.1%), with a smaller share in Quasi-Scientific (17.2%) and only 9.7% in Scientific. By Time 2, the Quasi-Scientific profile grew substantially to 51.8%, whereas the Scientific profile accounted for 26.6%, and Pre-Scientific dropped to 21.5%. At Time 3, the majority of students remained Quasi-Scientific (58.2%), whereas Scientific dipped to 12% and Pre-Scientific rose to 28.9%. Finally, at Time 4, the Scientific profile rebounded to 31%, Quasi-Scientific fell to 39.8%, and Pre-Scientific remained stable at 29.3%.

Table 4
Results From the Final Latent Transition Solution (Dispersion Similarity)

Indicators	Profile 1 Mean [CI]	Profile 2 Mean [CI]	Profile 3 Mean [CI]
Evaluation	1.56 [1.42, 1.66]	1.82 [1.64, 2.04]	1.80 [1.68, 1.96]
Preplausibility	−.690 [−1.49, .48]	4.97 [3.50, 6.65]	2.85 [2.10, 3.73]
Postplausibility	.694 [.051, 1.57]	5.50 [3.49, 7.37]	3.11 [2.58, 3.90]
Preknowledge	3.37 [3.28, 3.44]	3.95 [3.70, 4.31]	3.44 [3.26, 3.67]
Postknowledge	3.45 [3.31, 3.55]	4.17 [3.91, 4.53]	3.57 [3.35, 3.86]
	Variance [CI]	Variance [CI]	Variance [CI]
Evaluation	.205 [.136, .259]	.313 [.136, .456]	.218 [.116, .340]
Preplausibility	12.4 [9.63, 15.61]	3.34 [1.75, 5.34]	2.70 [1.75, 3.88]
Postplausibility	12.5 [9.22, 15.6]	3.72 [.477, 8.23]	2.22 [.379, 5.14]
Preknowledge	.151 [.108, .188]	.133 [.048, .183]	.114 [.070, .152]
Postknowledge	.250 [.172, .324]	.153 [.068, .204]	.135 [.082, .173]

Note. CI = 95% confidence interval.

Latent Transitions

To examine how students transitioned among the three profiles over time, we converted the final dispersion model from the similarity tests into a LTA with class regressions (see Table 6 and Figure 5). Across time points, Quasi-Scientific (Profile 3) showed the greatest stability at Times 2 (53.1%) and 3 (62.5%), whereas by Time 4, Scientific (Profile 2) was nearly fully retained (97.5%). In contrast, membership in the Scientific profile was notably less stable at earlier time points, with only 27.6% retention at Time 2 and 26.3% at Time 3. Similarly, Pre-Scientific (Profile 1) exhibited lower stability at Time 2 (21.7%) before showing moderate stability at Time 3 (53.1%) and a decline again at Time 4 (42.0%). Taken together, profiles marked by lower or initially negative plausibility judgments (namely, Pre-Scientific) tended to show more volatility over time, whereas higher-plausibility profiles eventually became relatively stable.

Table 5
Final Class Counts and Proportions for Each Latent Profile Across Each Time Point

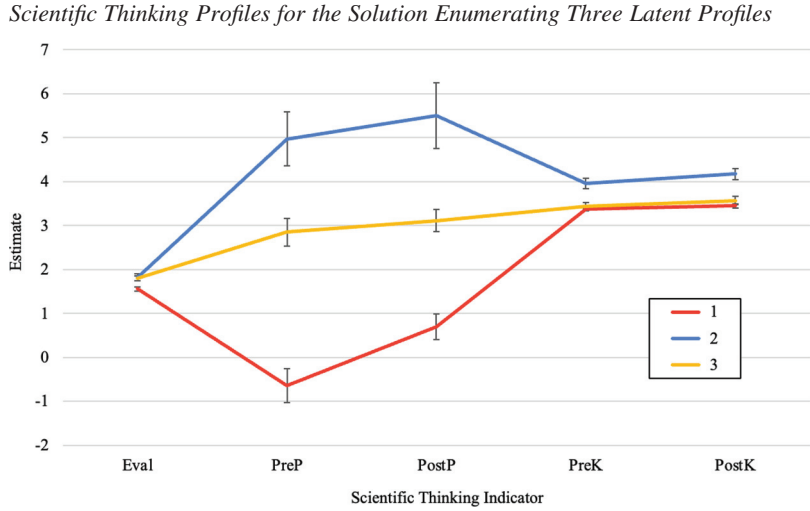
Time	Pre-Scientific (Profile 1) % (*n*)	Scientific (Profile 2) % (*n*)	Quasi-Scientific (Profile 3) % (*n*)
Time 1	73.1 (143)	9.74 (19)	17.2 (34)
Time 2	21.5 (42)	26.6 (52)	51.8 (102)
Time 3	29.8 (58)	12.0 (24)	58.2 (114)
Time 4	29.4 (57)	30.9 (61)	39.8 (78)

In terms of profile-to-profile transitions, it was less common for students in Quasi-Scientific or Scientific to move into Pre-Scientific. By contrast, those who began in Pre-Scientific frequently shifted toward Quasi-Scientific (52% at Time 2; 45.7% at Time 3; and 47.4% at Time 4), reflecting the comparatively lower likelihood of remaining in the Pre-Scientific group across time. Meanwhile, transitions out of Scientific occurred primarily into Quasi-Scientific (48.3% at Time 2; 60.7% at Time 3), though such movement became negligible by Time 4. Overall, these patterns suggest that while Pre-Scientific and Scientific memberships may be relatively fluid at earlier time points, learners tend to converge in Quasi-Scientific in the short term and eventually stabilize in Scientific as they continue through the instructional sequence. Also notable is the persistence of a small subset of students in the Pre-Scientific group, and the infrequency of transitions directly from the Pre-Scientific to the Scientific profile across consecutive time points.

Predictors of Profile Membership

Predictors were then added to our LTA model of dispersion similarity. We first estimated a model where predictors were freely

Figure 3
Scientific Thinking Profiles for the Solution Enumerating Three Latent Profiles



Note. Error bars represent ±1 standard error. Eval = evaluation; PreP = preplausibility; PostP = postplausibility; PreK = preknowledge; PostK = postknowledge. See the online article for the color version of this figure.

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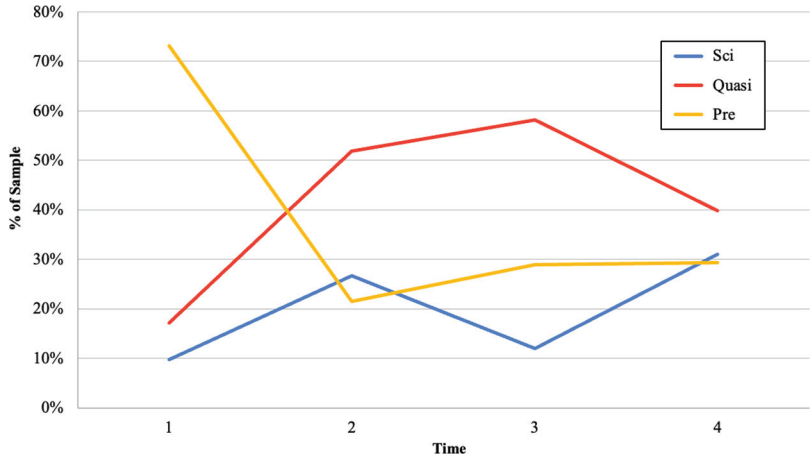
Figure 4
Proportions of Sample in Each Latent Profile Across Four Timepoints



Note. Sci = Scientific profile; Quasi = Quasi-Scientific profile; Pre = Pre-Scientific profile. See the online article for the color version of this figure.

estimated at Time 1 profiles and across time points. We contrasted this model with one in which predictors were estimated freely across all time points only and one where the predictor's estimates were constrained to be equal across time points and profiles. As shown in Table 2, we found that the model of predictive similarity had the lowest values for CAIC and BIC when compared with the alternative models, supporting the predictive similarity of our model. The findings from the logistic regressions are displayed in Table 3.

Across all three predictors, none of the estimates reached statistical significance in distinguishing Pre-Scientific or Scientific status relative to Quasi-Scientific (all $p > .05$). Specifically, neither gender nor region showed meaningful differences in the likelihood of belonging to the Pre-Scientific or Scientific groups, and the effect of age, while trending positive for the Scientific profile, did not achieve significance. Taken together, these results indicate that demographic factors did not account for the variability in students' scientific thinking profiles. When evaluating MEL order, we found that whether an observation came from the first, second, third, or fourth MEL activity significantly influenced profile membership only at Time 1 (see the LTA three-class model output in the supplemental materials).

Table 6
Transition Probabilities for the Final Latent Transition Model

Transitions	Latent profiles		
	Pre-Scientific	Scientific	Quasi-Scientific
Time 1 (rows) to Time 2 (columns):			
Pre-Scientific	**0.217**	0.263	0.520
Scientific	0.240	**0.276**	0.483
Quasi-Scientific	0.194	0.275	**0.531**
Time 2 (rows) to Time 3 (columns):			
Pre-Scientific	**0.531**	0.013	0.457
Scientific	0.130	**0.263**	0.607
Quasi-Scientific	0.287	0.088	**0.625**
Time 3 (rows) to Time 4 (columns):			
Pre-Scientific	**0.420**	0.106	0.474
Scientific	0.001	**0.975**	0.023
Quasi-Scientific	0.287	0.277	**0.436**

Note. Bolded numbers represent the within-person stability (i.e., individuals whose profile memberships did not change in the transition).

Discussion

Our study identified discernible scientific thinking profiles in the context of middle and high school science classroom learning. Specifically, we investigated profiles emerging around three constructs used to characterize scientific thinking: knowledge about scientific topics of social relevance, scientific evaluations about the connections between evidence and explanations of relevant phenomena, and judgments about the plausibility of scientific explanations and alternative, nonscientific explanations. Given that such scientific thinking is often a challenge for students, we employed the MEL scaffold, which prior research has shown to be effective in naturalistic (i.e., ecologically valid) classroom learning situations (for a summary, see Lombardi & Bailey, 2024). However, these prior studies were exclusively variable-centered, which may limit the situational nature of the findings, as well as the ability to provide adaptable and flexible guidance for instruction. This study advanced prior research on MEL scaffolds by employing a person-centered approach. By identifying distinct profiles of scientific thinking, this study offered a more nuanced understanding of how students engaged with and learned about scientific topics of social relevance. This approach provided not only empirical insights but also practical implications for designing differentiated instructional strategies in science classrooms (Howard & Hoffman, 2018).

The methodological approach of the study was robust, involving year five data (the last year of data collection) in a 6-year design-based research project. The research team was interdisciplinary, including classroom teachers, science education researchers, educational psychologists, and scientists. This added a rich diversity of perspectives to the study design and analysis. The sample included students from different regions in the United States, providing a broad demographic representation. The analytical

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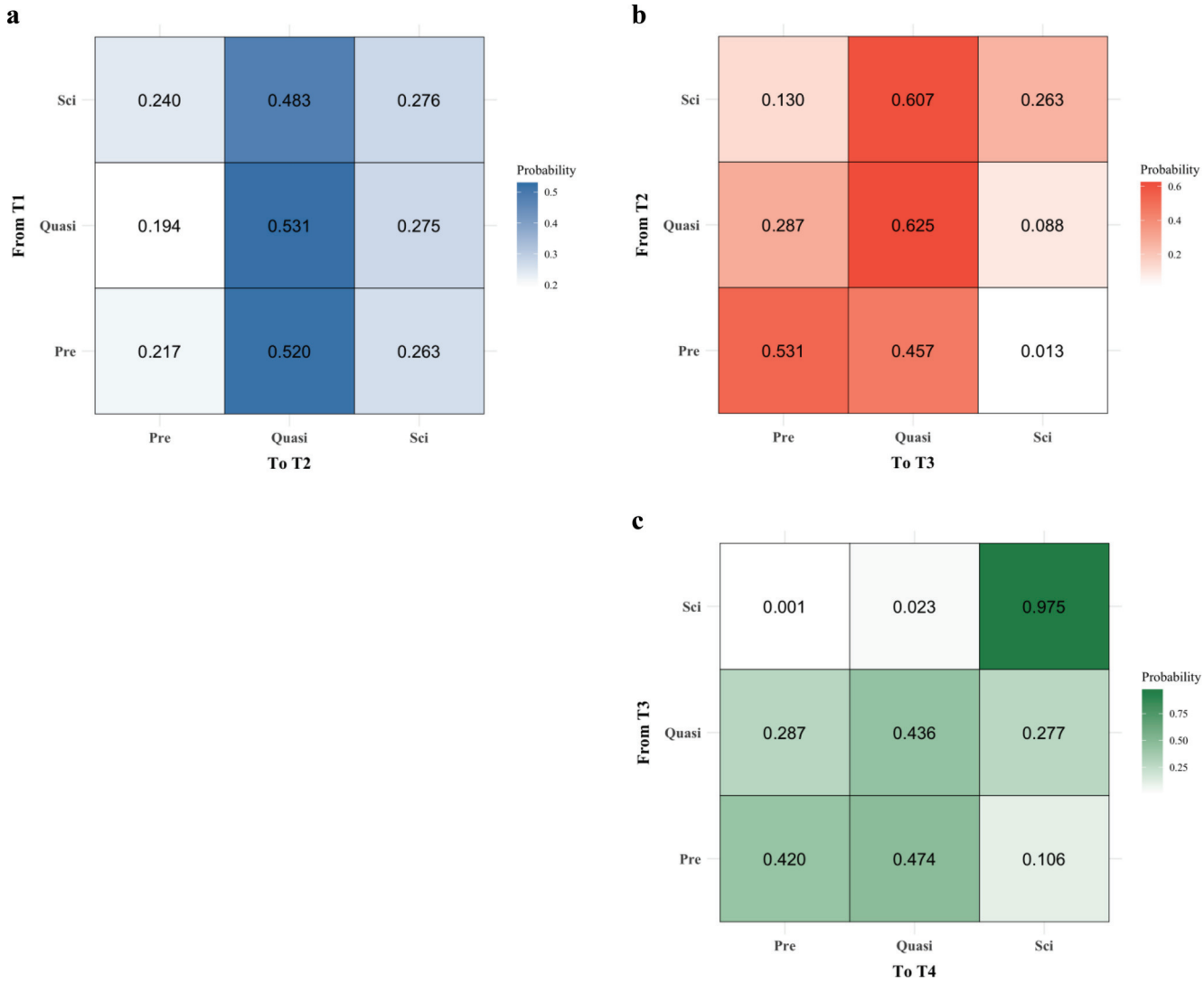
Figure 5

Transition Probabilities Between Latent Profiles Across Four Timepoints



Note. (a) Transition probabilities from Time 1 to Time 2; (b) transition probabilities from Time 2 to Time 3; (c) transition probabilities from Time 3 to Time 4. Sci = Scientific profile; Quasi = Quasi-Scientific profile; Pre = Pre-Scientific profile. See the online article for the color version of this figure.

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approach employed, LTA, identified common groupings of students based on knowledge, levels of evaluation, and plausibility judgments, as well as transitions between those groupings across successive scaffolded science lessons.

Our findings reveal three distinct profiles of scientific thinking—Scientific, Quasi-Scientific, and Pre-Scientific—each representing a different orientation toward scientific explanations. Students in the Scientific profile begin instruction with relatively high plausibility judgments favoring scientific explanations and greater knowledge of scientific topics of social relevance. Those in the Quasi-Scientific profile demonstrate strong evaluative skills but may still weigh alternative explanations with some epistemic uncertainty, suggesting an intermediary stage where additional instructional support could reinforce conceptual understanding. In contrast, students in the Pre-Scientific profile initially perceive nonscientific explanations as more plausible but show potential for shifts in this stance when provided with scaffolded support.

Theoretically, these profiles appear to align with epistemic stances identified in prior empirical and theoretical work. The Pre-Scientific profile resembles a prereflective stance (absolutist or multiplist orientation), in which knowledge is treated as factual or a matter of opinion, with limited reliance on shared standards for adjudicating claims. The Quasi-Scientific profile reflects a quasireflective stance (emergent evaluativist orientation), where learners may not consistently prefer a scientific account and instead vary depending on topic and context. The Scientific profile reflects a reflective stance (evaluativist orientation), marked by explicit coordination of evidence and explanations along with greater endorsement of the scientific model (King & Kitchener, 2004; Kuhn et al., 2000; Kuhn, 2010b). In this way, our person-

centered findings provide a profile-level expression of these classic stage-like accounts, while also showing how stance and performance can diverge in classroom activity.

The Scientific profile was characterized by the highest evaluation scores and the highest pre- and postinstruction plausibility judgments. The Scientific profile also showed knowledge gains, though more modest than the other two profiles, perhaps due to a higher baseline. Students in the Pre-Scientific profile had the lowest evaluation and knowledge scores and initially endorsed nonscientific explanations. However, they exhibit the largest plausibility shifts after engaging with scaffolded instruction, suggesting that explicit, evidence-focused activities can facilitate science learning. The Quasi-Scientific profile falls between the other two, with students demonstrating fairly strong evaluation skills but holding moderate plausibility shifts and knowledge gains. These learners appear to evaluate evidence effectively but may require additional reinforcement to consolidate their scientific thinking. The dynamic nature of these transitions underscores the critical role of instructional scaffolding in supporting students at varying levels of their learning trajectory.

The shifting profile proportions (see Table 5) underscore the dynamic nature of students' scientific thinking, particularly in terms of plausibility judgments. Early in the instructional sequence (Time 1), the overwhelming majority of students fell into the Pre-Scientific profile, suggesting that many students tend to favor nonscientific explanations. By Time 2, a notable share of these learners moved into the Quasi-Scientific profile—indicating they had developed stronger evaluative skills and plausibility judgments after some initial instruction—while a smaller but still substantial subset advanced directly to the Scientific profile. At Time 3, however, the proportion of the Scientific profile declined again, perhaps reflecting the introduction of new, more challenging content that prompted students to revert or recalibrate their thinking (and thus revert to a less scientific stance). Interestingly, by Time 4, the Scientific group rebounded and Quasi-Scientific participants remained a plurality, suggesting that repeated or cumulative scaffolding experiences eventually stabilized some learners' epistemic stances in favor of the scientific explanation, whereas others maintained intermediate levels of knowledge and plausibility judgments.

This pattern comports with knowledge reconstruction models in which learners often traverse intermediate states before stabilizing on a new conception (Dole & Sinatra, 1998; Kendeou, 2024; Lombardi et al., 2016). The prominence and mid-course stability of the Quasi-Scientific profile suggest it may function as a "transitional attractor," where evaluative resources are active but stance-level endorsement remains unsettled. Consistent with overlapping-waves views of development, we also observe temporary backshifts as new content raises uncertainty, followed by renewed consolidation with continued scaffolded practice (Siegler & Jenkins, 2014; Kuhn, 1995).

Looking within the transition probabilities reveals that learners' epistemic stances are not static but evolve as they progress through repeated instructional episodes. The Quasi-Scientific profile exhibited the greatest stability around midinstruction (Times 2 and 3), with more than half of its members remaining in that profile across those time points. This finding suggests that moderately high evaluation skills and plausibility judgments—hallmarks of the Quasi-Scientific group—can provide a relatively secure cognitive "landing spot" once students have moved beyond the more naive, nonscientific orientation (i.e., Pre-Scientific). By contrast, individuals in the Scientific group were much less stable early on, possibly because advanced scientific thinking (i.e., high plausibility for the scientific explanation, relatively higher knowledge) might require repeated reinforcement to consolidate. Indeed, it was only by Time 4 that Scientific membership became almost universally retained (97.5%), indicating that over the long term, recurrent exposure to evidence-based activities may solidify an evaluativist stance that becomes resistant to backsliding.

Viewed through a dynamic perspective, these transitions echo prior latent transition work in science learning that finds moderate stability in mid-level profiles and delayed consolidation of the most advanced profiles (e.g., Gillet et al., 2017; Schlatter et al., 2021). The short-term convergence on Quasi-Scientific followed by later retention in Scientific may indicate that repeated opportunities to apply evaluation criteria can gradually shift both performance (how students evaluate) and/or stance (how plausible the scientific account seems), rather than producing an immediate step-change.

Interestingly, Pre-Scientific membership underwent fluctuations as well, showing low stability at Time 2 (24.6%) but moderate stability at Time 3 (60.4%), before dropping again at Time 4. This pattern reinforces the notion that learners who initially lean on nonscientific explanations are more likely to transition out of that stance once they gain some grounding in evaluation skills and build their domain knowledge. In particular, many Pre-Scientific learners moved into the Quasi-Scientific profile, confirming that, for some students, the development of scientific thinking is a multistep process rather than an immediate leap to a fully Scientific position. Still, a small subset of students remained Pre-Scientific into later time points, underscoring that instructional interventions must account for varying degrees of epistemic and conceptual malleability. Overall, the latent transition findings align with a dynamic view of conceptual and epistemic change, where learners often pass through intermediate states (Quasi-Scientific) before ultimately stabilizing in a more scientifically aligned epistemic stance (Scientific)—provided they receive ongoing support designed to reinforce evidence-based reasoning.

Collectively, these results underscore the heterogeneity of learners' scientific thinking, especially as it pertains to plausibility judgments. In practice, this suggests not all students enter the classroom with the same epistemic stance, and the effectiveness of instruction can vary based on whether learners already endorse scientific models (Scientific), remain unconvinced (Pre-Scientific), or occupy a more in-between status (Quasi-Scientific). Recognizing these subgroups may help design instruction that is more responsive to students' prior knowledge and skills. For students in the Pre-Scientific profile, explicit scaffolding that emphasizes the iterative nature of scientific reasoning—such as structured refutational texts or dialogic argumentation (see, e.g., Danielson et al., 2025; Nussbaum, 2021)—may facilitate more robust epistemic shifts. For those in the Quasi-Scientific profile, reinforcing connections between evidence and explanatory models through guided inquiry or extended evaluation tasks may enhance their confidence in scientific conclusions (Governor et al., 2025). Meanwhile, students in the Scientific profile could benefit from opportunities to engage in more advanced scientific thinking, such as evaluating the limitations of scientific models or exploring contested debates in science

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(Compton et al., 2021). By tailoring instruction to meet students where they are in their epistemic development, educators can more effectively foster scientific thinking and engagement across diverse learner populations.

We should note that—given the novelty of this study—norms for effect size expectation and interpretation do not readily exist in the extant literature. Therefore, we followed the guidance from Giner-Sorolla and colleagues (2019) and interpreted our findings in terms of *practical* effects to establish meaning. Our past variable-centered studies have consistently shown 8%–10% science learning gains across various populations of middle and high school students using the MEL scaffold. This is an appreciable gain for just a single 90-min activity, a claim anecdotally endorsed by the education practitioners who have collaborated in this line of research over the years. The findings from the present study's person-centered analysis suggest more fine-grained distinctions about subgroups of middle and high school students, which holds a promise for informing more specific teaching practices that facilitate instructional differentiation and optimize student learning. In terms of approaching such practically significant effects (i.e., via using cost-benefit analysis as suggested by Kraft, 2020), the relative cost of employing scaffolding like the MEL diagrams, which are of low implementation cost and can easily replace less effective classroom strategies (e.g., an exclusive focus on lecture; Lombardi et al., 2021) may be of particular benefit in helping shift some student plausibility judgments toward a more scientific stance. At the same time, effective scaffolding designed to promote scientific thinking can still support learning among student subgroups where plausibility judgments already reflect a more scientific stance before instruction.

The practical meaning of the profiles as enumerated by our accepted solution comes into clearer relief when considering the transition probabilities between the profiles at each time point. Our findings show that students exhibited more refined scientific thinking and greater acceptance of scientific explanations over the period in which they engaged with multiple MEL diagrams. This progression underscores the value of repeated and varied engagement with scientific concepts, supporting the idea that continuous engagement in scientifically structured tasks enhances students' scientific thinking. It may also be the case that earlier interactions with a novel instructional scaffold entail elevated cognitive load on the part of the learner (Davis, 2000). Students may automatize the "procedural rules" of the MEL diagrams over repeated exposures allowing them to, for example, expend less cognitive effort remembering which arrow to use for which link weight, and instead engage more deeply about the relationship between evidence and explanation. This potentially allows students to internalize the scaffold with repeated instructional exposure and develop a scientific habit of mind that may transfer beyond the classroom context (Darling-Hammond et al., 2020).

The study has several implications for educational practice and future research. First, it underscores the importance of considering individual differences in students' scientific thinking. We suggest that educational interventions and curricula pay more attention to person-centered research when designing instructional tools, with such tools catering to differing profiles of students to maximize learning outcomes. The findings highlight the potential of scaffolded learning tools, such as MEL diagrams, to be associated with improvements in students' understanding of scientific concepts and reasoning skills. However, we should note that the MEL scaffold is just one of a small body of burgeoning instructional methods designed to facilitate students' evaluations of the connections between scientific explanations and alternative explanations of phenomena (see, e.g., Darner, 2019; Dauer et al., 2022; Jimenez et al., 2024). Educators and curriculum developers should consider student profiles when designing instructional materials and interventions, ensuring that they cater to the diverse needs of students. For instance, students with lower plausibility scores may benefit from more frequent and explicit discussions about the nature of scientific evidence and its role in shaping scientific consensus.

Limitations, Future Directions, and Concluding Thoughts

Although our methods for addressing missing data are standard in research (Hertel, 1976; Marini et al., 1980), they may introduce some level of bias or affect the representativeness of the sample. For example, our data come from the 2021–2022 academic school year, a period when many schools, teachers, and students were readjusting to in-person instruction post-COVID-19. Evidence already exists to suggest that the negative effects of the pandemic were not evenly distributed among student groups (Ariyo et al., 2022; Betthäuser et al., 2023; Hammerstein et al., 2021). If COVID-19-related experiences (e.g., absenteeism) nonrandomly influenced our missing data, this could introduce bias. The use of LTA, although effective, is subject to the challenge of determining the "correct" number of profiles, a common issue in mixture modeling (Nylund et al., 2007). The decision to choose a three-profile solution, although theoretically and statistically justified, might oversimplify the complexity of student learning profiles in scientific reasoning. Internal consistency for the topic-specific knowledge questionnaires was uneven and warrants consideration. Although reliability for three out of the four topics met or approached conventional thresholds, the cosmic origins knowledge survey in particular showed lower reliability ($\lambda_2 = .49$). Such modest reliability can attenuate effect sizes and increase measurement error; therefore, conclusions that hinge on knowledge gains should be interpreted with appropriate caution.

Although impossible to implement due to demands on participating teachers' by curricular sequencing, a fully crossed design that rotates scaffold *type* × *topic* × *time* would provide a more stringent test of generalizability. Because the study did not include a business-as-usual comparison group, improvements cannot be attributed unequivocally to the MEL scaffolds. Although the effectiveness of MELs compared with other instructional scaffolds has been previously demonstrated (Lombardi, Bailey, et al., 2018), the findings reported here should not be interpreted as causal. In addition, given that we found MEL order did influence some students' profile memberships, future research should consider employing a fully counterbalanced design to better delineate learning trajectories within and between students over time.

These limitations were considered and addressed as fully as the study design allowed. The analytic methods employed, including the use of FIML and robust model verification procedures, bolster the result's reliability and meaning. In addition, conclusions drawn are based on comprehensive data analysis and are supported by the consistency of the findings across

different tests and comparisons and are parsimonious through the lens of the theoretical framework.

We interpreted the meaning of the study's findings in terms of practical effects. The study also warrants attention for its *basic practical effects*, which "depend on a paradigm's ability to be replicated, extended, and inspire further research" (Giner-Sorolla et al., 2019, p. 9). Our LTA not only delineated distinct profiles of scientific thinking and transitions between but also showcased the robustness of our analytical approach, underscoring its potential for wider application. These profiles, reflecting different levels of engagement with and understanding of scientific concepts, suggest that our study's framework could be effectively replicated and extended to various educational settings. For instance, educators in a different region or school system could apply a similar mixture modeling approach to identify student profiles in their own context, adapting instructional strategies to better suit the identified needs. Furthermore, the adaptability of our method paves the way for its application across diverse educational landscapes, potentially improving the science learning of varied student populations. This scalability and adaptability are crucial for advancing educational research and practice, particularly in enhancing students' scientific reasoning and understanding. In addition, the insights gained from these profiles could inspire subsequent research, such as developing targeted instructional interventions that are precisely tailored to different learner profiles. For example, if one profile shows students with strong plausibility judgments but lower knowledge scores, interventions could focus on enhancing factual understanding while leveraging their existing evaluative skills. Such a targeted approach could lead to more effective educational strategies that are specifically designed to maximize learning outcomes based on individual student needs and characteristics.

The contour of the Pre-Scientific profile and its changing composition across time provides a compelling avenue for further research. This profile included students who, despite mostly accurate prior knowledge, showed lower acceptance of scientific explanations. This suggests a potential paradox within the scientific thinking of such students; that is, the presence of accurate knowledge of a scientific phenomenon alongside doubt around the scientific account for the phenomenon, akin to that posited by Kendeou (2024) in their *Theory of Knowledge Revision*. It is possible, for example, that a student's working explanatory model for a given scientific phenomenon may not be represented by the options presented within the MEL activity. If this were the case, plausibility as measured in this study may not capture an individual's true epistemic orientation to the scientific consensus. Further work is needed to understand the nature of this dynamic. Doing so may reveal opportunities for practitioners to better engage students' epistemic judgments (i.e., plausibility) in the service of science learning. On the theoretical and philosophical accounts, informing on this matter may further the debate about whether (science) "knowledge is merely true belief" (Myers-Schulz & Schwitzgebel, 2013; Sartwell, 1991).

It is also worth noting that our analysis found little separation between profiles in terms of students' levels of evaluation of explanatory models. One plausible explanation is the narrow range of variance in how students generated and recorded their evaluative judgments. Our measure required students to write brief rationales for only a subset of their evidence-model links—an approach that may capture minimal variance if, for example, most students can articulate only a basic (i.e., descriptive rather than critical) connection between evidence and explanation. Alternatively, it is possible that the instructional sequence itself (e.g., repeated practice with the same scaffolding) helped learners acquire a minimum threshold of evaluation skills, thereby making it less likely that the three profiles would separate on this measure. Theoretically, a threshold effect is also plausible; once instruction socializes a minimal common set of epistemic criteria for linking evidence and models, variance in evaluation narrows even as stance-level plausibility remains more labile (Osborne, 2010; Nussbaum, 2011). Future work involving more fine-grained assessments and/or targeted instruction on constructing critical evaluations could help detect subtler or more continuous differences in students' evaluative processes.

In summary, we suggest that future research focus on further exploring the nuances of these and other profiles and how they interact with different instructional strategies. Expanding the demographic and geographic diversity of the sample could also enhance the generalizability of the findings and provide a more comprehensive understanding of student learning profiles in scientific reasoning. Our logistic regression did not find regional or demographic factors to significantly predict profile membership, consistent with other research suggesting differences in sociodemographic categories do not significantly relate to variations in scientific thinking and reasoning (Lazonder et al., 2021). These results, however, should be interpreted with caution. It may be the case that group imbalances within our sample restricted our ability to detect potential effects of demographic variables. It may also be the case that unitary, census-like categories do not capture the proximate experiences that more meaningfully and significantly influence an individual's learning and development (Cole, 2009; Gutiérrez & Rogoff, 2003). Researchers should continue to consider and investigate how individual contexts interact with scientific thinking and its development.

Scientific topics of social relevance represent fertile ground for developing students' scientific thinking. In particular, with such topics often being controversial, it is important for students to practice the skills of knowledge construction and plausibility appraisal when considering the connections between evidence and alternative explanations of phenomena that impact their local, regional, and global communities. The present study contributes to a program of research that is committed to equipping students with the tools and agency to become problem solvers, now and in the future, who will benefit their communities by taking reasoned, scientific, and civically-responsible action. This study lays a groundwork for more personalized and effective science instructional strategies that support diverse scientific thinking and promote equitable solutions to shared challenges.

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